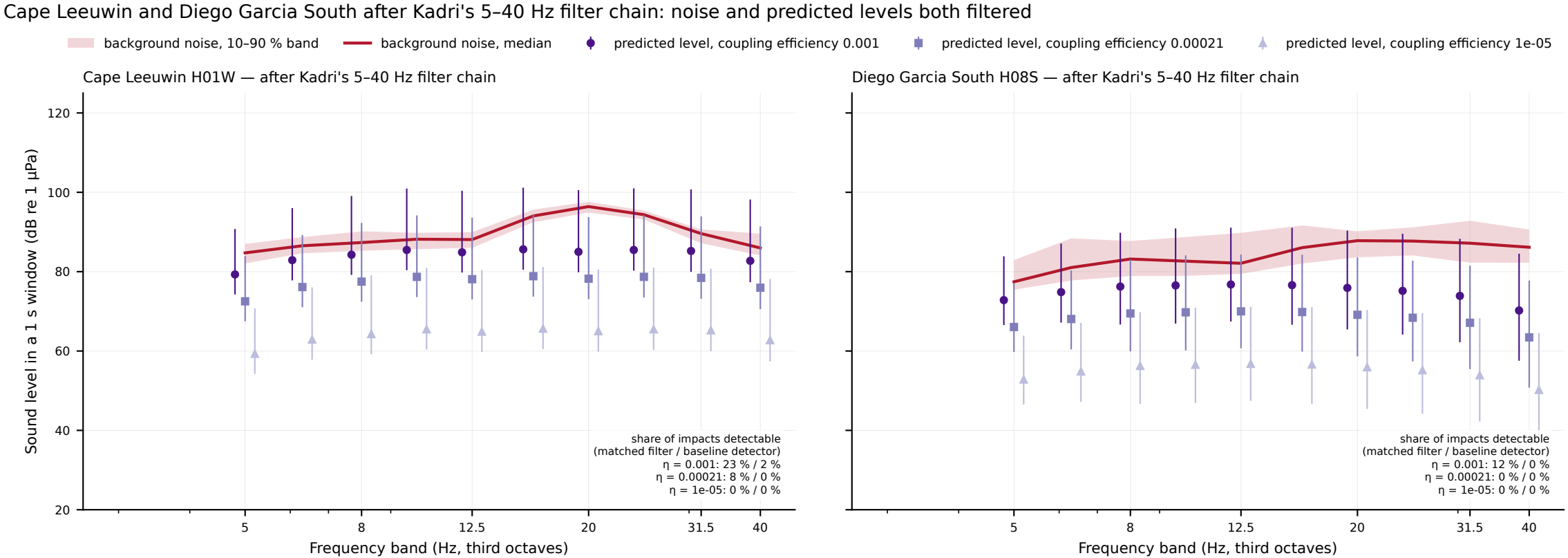

Plain-language note: The red curve is the background noise at the station, by frequency band; the shaded band holds 80 % of the one-second noise levels. The markers show the sound level that the simulated impacts would make at the station (median, with a bar from the 10th to the 90th percentile), for three assumed values of the share of impact energy that becomes sound. Where a marker sits above the red curve the impact would stand out in that band. The box gives the share of simulated impacts whose total 5–40 Hz level would be detected by the two detectors tested on real recordings. The 2–5 Hz band shows noise only: no sound-loss model exists there yet. Here both the noise and the predicted levels have been passed through the same 5 Hz high-pass and 2–40 Hz band-pass filters that Kadri (2017, 2019, 2024) used.

Technical note: Source posterior: end of flight run J (74 seeds, 313.6 M rows) + northern supplement; weighted subsample 2 M draws per option x p; prereg run_j_levels.py 4277320. Option 00:19 held out (D0'), dive probability p = 0.5, eta in {1e-3, 2.1e-4, 1e-5}; E = contact kinetic energy, tau = tau90 (k = 1), 10 m source, f^-2 one-pole spectrum; TL from the whole-arc KRAKEN grid (offset-0 line, RAM-corrected at the loggers, RAM delta at H01W/H08S, none at H08N/3250), blocked bands zero. Levels are 1 s-equivalent third-octave band SPL (T_eff = 2.68 s), dB re 1 uPa. Noise: IMOS third-octave 1 s percentiles (14-day background set for 3376/3274/3250, 8 Mar 2014 role-B recordings for 3315/3275); IMS = Blackman App. B 2003 PSD x band width (digitised). Detectable = SNR on the 5-40 Hz total >= 7 dB (envelope matched filter, P_D = 0.5, alpha 0.01) or >= 15 dB (STA/LTA), model-matched upper bounds from the weak-signal trial on 3376 background; SNR uses the module's 5-40 Hz noise medians (imos2b; 3376 proxy + Blackman offset at IMS, +4.8 dB triad gain). PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, now a source-term caveat after the EW0112 result) and C2 (coupling efficiency not calibrated; F-35A upper anchor 2.1e-4). Kadri chain: 4th-order Butterworth 5 Hz high-pass x 2-40 Hz band-pass (order assumed), each band's energy scaled by the mean |H|^2 over the band; the 'detectable' box is unchanged (it is on the unfiltered 5-40 Hz total).